

Sri Datta Ganesh Bandreddi

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EDUCATION

Carnegie Mellon University

M.S. in AI Engineering (Biomedical Engineering) | GPA: 4.00/4.00

Pittsburgh, PA
Expected Dec 2026

- **Coursework:** Introduction to Deep Learning (24-788), Learning for 3D Vision, and AI and IoT.
- **Teaching Assistant** for AI and IoT; built the teaching material, including an IoT VLM Colab notebook.

Visvesvaraya National Institute of Technology (VNIT)

B.Tech, Mechanical Engineering | GPA: 8.44/10.00 (Scholaro-scaled 3.63/4.00)

Nagpur, India
Jul 2022

- **Coursework:** AI in Manufacturing, Industrial Robotics, Manufacturing Process Automation, Computer Graphics & Solid Modelling, Data Structures, Finite Element Method, Biomechanics, and Machine Design.

SKILLS

Languages: Python, C++, Go, Java, SQL, Bash / Shell scripting

GPU / Acceleration: CUDA, cuDNN, cuBLAS, TensorRT, Triton Inference Server, NCCL, Nsight Systems/Compute

LLM / Agents: LangGraph, LangChain, RAG, agentic tool-calling, FastAPI (serving)

DL Frameworks: PyTorch, JAX, TensorFlow, Hugging Face, NeMo / Megatron-LM, LoRA / QLoRA, FP8 / BF16 mixed precision

Data & Big Data: RAPIDS (cuDF, cuML), NVTabular, DALI, NumPy, SciPy, Pandas, SparkML, Ray, Apache Spark, H3, medallion architecture

Systems & MLOps / IaC: Docker, Kubernetes, Slurm, Terraform, Bazel, Jenkins / GitHub Actions, Helm, MLflow, Git, SonarQube

Cloud: AWS (ECS, SageMaker, S3, RDS, KMS, EventBridge), GCP

Databases: PostgreSQL, Citus, Neo4j, memgraph, JanusGraph, MongoDB, Cassandra, ScyllaDB, Redis, Valkey, Kafka, Iceberg, Elasticsearch, FAISS

Security & Observability: Prometheus, OpenReplay, AWS KMS / encryption-at-rest, BYOK, GDPR, SAML / SSO / RBAC / OAuth

Frontend: ReactJS, Next.js

PROFESSIONAL EXPERIENCE

Motional

Autonomous-vehicle data platform

Data Platform Intern (Infrastructure)

May 2026 – Aug 2026

- Designed, prototyped, and shipped a metadata-extraction system so teams could learn what is inside a drivelog (about 1 TB each, petabytes in total) before paying to extract the whole thing. Delivered a working PoC in staging and authored the production design.
- Built the production system on a **medallion architecture** so it could grow into a reusable, first-class data product.
- Enriched drivetlogs with weather and map layers, then used **H3 geospatial indexing** to roll up metrics across regions.
- Made scene-metadata retrieval fast with a new geospatial indexing scheme, and indexed scenes as embeddings for similarity search. Filed a **provisional patent** on the approach.
- Added an **agentic AI layer (LangGraph)** that lets people find drivetlogs in plain English through tool calls, served behind a **FastAPI** backend.
- Benchmarked **Ray Data against Spark** on Kubernetes and went with Ray, wiring it up with AWS EventBridge, S3, and RDS.

Phenom

Hyderabad, India

Product Development Engineer, Enterprise Talent Graph

Dec 2023 – Jul 2025

- Led six engineers to re-architect a multi-tenant AI analytics platform, refactoring shared services and cutting cross-tenant coupling so it held up under heavy concurrent load.
- Grew a production knowledge graph past 1M entities with continuous ingestion, designing validated aggregation pipelines on top of Neo4j schemas and Iceberg versioned storage.
- Kept workloads highly available for 100+ enterprise clients, running distributed Kafka, PostgreSQL, and Redis clusters and building CI/CD with Jenkins and Helm on Kubernetes.
- Sped up AI-driven query responses by 40% across 100+ clients by tuning PostgreSQL horizontal sharding.
- Owned the entire Kafka platform: wrote a custom exporter (shell scripting plus Prometheus scraping) and stood up Kafka UI, giving the team live visibility into partitioning, rebalancing, consumer lag, and node failures, and led the migration from ZooKeeper to **KRaft**.
- Pioneered running Redis and other databases on Kubernetes for elastic scaling, and owned the Helm charts that kept those deploys repeatable.
- Worked on the central crypto team, implementing encryption at rest and owning key management and rotation through AWS KMS, and helped build **Bring-Your-Own-Key (BYOK)** support for EU customers.
- Enforced **GDPR data-retention** policies across every data product, safeguarding PII and guaranteeing deletion.

- Rewrote a proxy service from scratch in **Go**, cutting startup time and lifting throughput by 30%, and set up Docker Compose integration-testing pipelines.
- Benchmarked ScyllaDB, Cassandra, PostgreSQL, and Citus, built RAG pipelines over related graph entities, ran OpenReplay for UI-behavior heatmaps, and negotiated long-term Neo4j licensing.
- *Recognition*: won the Phenomenal Award for Outstanding Performance in both Q1 and Q2 of 2024.

BYJU'S

Hyderabad, India

Software Engineer, Core Platform Team

Jul 2022 – Dec 2023

- Cut infrastructure costs by 70% while keeping 1M+ users well served, deploying scheduled AWS ECS auto-scaling (Terraform IaC) tuned to real traffic patterns.
- Halved reliance on external identity services by leading development of an in-house authentication and authorization platform in Go and Java (Spring Boot).
- Drove an internal **SSO** project end to end, implementing SAML, SSO, and RBAC so every service call was authenticated, authorized, and traceable, while mentoring and leading interns.
- Built a **multi-channel communications platform** with real-time targeting that tracked exactly how many messages went out, which let us reconcile charges to **save at least \$100K** and **lift user engagement by 20%**.
- Load-tested target APIs and **cut their latency by 30%**, and built an in-house CRM backend in Java (Spring Boot) that shipped in two months.
- Held admin on 5 repositories as a fresher and owned the full auth and comms flow, and made it through 7 rounds of layoffs as the company wound down, a good sign of how critical the work was.

Phenom

Hyderabad, India

Machine Learning Intern

May 2021 – Jun 2022

- Lifted recommendation accuracy by building persona-classification models with spaCy and BERT embeddings across 3 data sources, feeding personalized content into production recommendation pipelines.
- Made recommender preprocessing much faster by moving CPU pipelines to NVTabular on GPU, cutting latency and proving the scale-up across 2 production databases.

PUBLICATIONS & RESEARCH

- **FLARE: Fast Low-rank Attention Routing Engine.** V. Puri, A. Joglekar, **S. D. G. Bandreddi**, K. Ferguson, Y. Chen, Y. J. Zhang, L. B. Kara. arXiv:2508.12594; submitted to TMLR (*3rd author*). Later picked up by an NVIDIA and General Motors crash-dynamics effort (arXiv:2605.27758), which dropped a FLARE-based backbone into GeoTransolver on NVIDIA PhysicsNeMo for roughly 2x lower memory.
- **FLARE++: Low-rank Attention with Dynamic Attention Routing.** arXiv:2608.11519, in progress and aimed at ICLR 2026 (*2nd author*). Sharpens how FLARE routes information through its latent tokens.
- **Comparative FEA Study of Palatal Expansion Devices.** **S. D. Bandreddi**, V. Vaidya, R. R. Salguti, R. Uddanwadiker. Presented at ICMAE 2022 in Bratislava, Slovakia (*first author*).

SELECTED PROJECTS

FLARE / Efficient Attention (CMU Research)

Aug 2025 – Present

- Showed over 200x speedups and sub-quadratic memory scaling for linear attention on Long Range Arena (1k to 8k tokens), and profiled multi-GPU NCCL training across FP8/BF16 with Nsight Systems on Slurm.
- Owned the LRA benchmark suite and the latent-expression ablations, and took FLARE into image classification and diffusion, building every architecture change and the GPU image data pipelines (NVIDIA DALI).

FLARE++, Dynamic Attention Routing (CMU Research)

Ongoing

- Ran token-mixer ablations (AB-UPT versus FLARE) on DrivAerML and PDEBench to test dynamic attention routing and pin down which pieces of the design actually drive the accuracy and efficiency gains.

CT-Transformer, Vision Transformers (CMU)

Sep 2025 – Dec 2025

- Hit 96.22% macro AUC-ROC on 15-class VinBigData chest X-ray classification, beating ResNet-50, EfficientNet-B3, and ViT baselines, with a 210M-parameter Swin Transformer and fused attention kernels.
- Sped up inference 3x over PyTorch eager by converting to TensorRT (FP16/INT8), speeding preprocessing with DALI, and serving on Triton with adaptive batching.

Distributed LLM Fine-Tuning with Megatron-LM (CMU)

Aug 2025 – Oct 2025

- Fine-tuned a 7B-parameter LLM to stable convergence with about 40% less memory on four H100 GPUs, using tensor and pipeline parallelism (NCCL), gradient checkpointing, and LoRA/QLoRA adapters.
- Pushed training throughput to 2.1x over naive data parallelism by tuning NCCL, gradient bucketing, and optimizer-state sharding.

EMG Hand Gesture Recognition (CMU)

Sep 2025 – Oct 2025

- Built PyTorch LSTM/CNN and attention models for EMG hand-gesture classification, reaching 85 to 92% accuracy on 16-channel, 1000 Hz signals, with controlled ablations over subject-wise splits.

GitAsk, Local-First RAG over GitHub Repos (Open Source)

- Core contributor to a browser-based tool that turns any repo into a chat assistant powered by an in-browser LLM, with no backend needed. Built the BM25 sparse retrieval for hybrid search, the embeddings (MiniLM), the security hardening (encrypted key vault), and the performance tuning.

AI-Driven Knowledge Graph for Obsidian (Personal)

Jul 2024 – Present

- Auto-links notes by pulling out entities and relationships with spaCy and BERT, with real-time graph clustering (NetworkX) and D3.js visualization.

Go Lopper (URL Shortener) & Trello Clone (Personal)

- Go Lopper is a URL shortener in Go that uses ULID keys in PostgreSQL for scalable inserts, with Redis caching and Docker deploys. Trello Clone is a Next.js and Appwrite to-do app with drag-and-drop, GPT-API task summaries, and Google auth.

Comparative FEA of Palatal Expansion Devices (B.Tech Capstone)

2022

- Ran an FEA comparison of MARPE and HYRAX devices, modeling skull stress and displacement with suture lines, and showed MARPE spreads stress better. Accepted for presentation at ICMAE 2022 (first author).

AWARDS & HONORS

- **Phenomenal Award for Outstanding Performance** at Phenom, Hyderabad (2024), for standout contributions in Q1 and Q2.
- **Perfect 4.00/4.00 GPA** at Carnegie Mellon University.
- **TartanHacks track winner** (early 2026) for a playable game about CMU that fit in under 15 KB total.
- **Provisional patent** (Motional, 2026) for geospatial indexing and retrieval of AV scene metadata.

LEADERSHIP & EXTRACURRICULAR

- **Student Mentor** at VNIT (2020 to 2022), guiding around 120 first-year Mechanical Engineering students.
- **Student Council Representative** at VNIT (2018 to 2022), carrying student feedback to faculty and improving course plans.
- Organizing Committee for Aarohi (VNIT cultural fest, 2018); active in the Cycling and Quiz clubs; district-level badminton (2019).
- **Languages:** English (Proficient), Hindi (Fluent), Telugu (Native), German (Basic).